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## Supplementary Material S4: Leave-One-Out Sensitivity Analysis for the Catheter-Failure Outcome

Leave-one-out (LOO) random-effects sensitivity analysis was performed on the four studies contributing to the pooled catheter-failure estimate (DerSimonian–Laird $\tau^2$, log-RR scale). Each row reports the pooled risk ratio (RR) and 95% confidence interval (CI) when the corresponding single study is removed from the model. The full-model pooled estimate (all four studies) was RR 1.225 (95% CI 0.995–1.508; $p = 0.056$; $I^2 = 0.0\%$).

**Table S4.**

*Leave-one-out (LOO) pooled risk ratios for the catheter-failure outcome (k = 4).*

| Study excluded | Weight in full model (%) | Pooled RR [95% CI] | p-value | $I^2$ (%) |
|---|---|---|---|---|
| **Kleidon (2025)** | 14.9% | 1.28 [1.02, 1.60] | 0.0316 | 0.0% |
| **Shokoohi (2019)** | 28.1% | 1.19 [0.93, 1.52] | 0.159 | 0.0% |
| **Leroux (2023)** | 7.2% | 1.23 [0.99, 1.52] | 0.0626 | 0.0% |
| **Saltarelli (2015)** | 49.8% | 1.18 [0.88, 1.58] | 0.277 | 0.0% |

**Footnotes:**

- CI, confidence interval; $I^2$, between-study heterogeneity statistic; LOO, leave-one-out; RR, risk ratio.
- Model: DerSimonian–Laird random-effects on log(RR); single-study removal.
- p-values are two-sided Wald-type tests on the pooled log-RR.
- Saltarelli (2015) carried 49.8% of the full-model weight; removing it attenuated the pooled estimate to RR 1.18 (95% CI 0.88–1.58, $p = 0.28$), indicating that the borderline catheter-failure signal depends substantially on this single largest study.
- All four contributing studies (Kleidon 2025, Shokoohi 2019, Leroux 2023, Saltarelli 2015) are reported in Table 1 of the main manuscript and in the PRISMA flow diagram.
- Full R analysis script: scripts/04_meta_analysis.R; underlying data: output/SupTable_LOO_catheter_failure.csv.